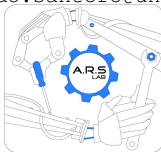


Basics of Artificial Intelligence

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Corso di "Introduction to Robotics and AI"

Artificial Intelligence and Autonomous Systems

Artificial Intelligence

Artificial Intelligence

The ability of an artificial system to behave in an intelligent way

Questions

- Behave: seem? show? mimic?
- Turing test
- Chinese Room (John Searle)
- What is the “intelligence”?
- Consciousness?

Autonomy

Autonomy

The ability of an artificial system to accomplish a goal without any external intervention

Questions & Thoughts

- **Is an autonomous system also intelligent?**
- An air conditioner is autonomous, is it intelligent?
- The Tesla car is autonomous, is it intelligent?
- An airplane is autonomous, is it intelligent?
- A Waking-up is autonomous, is it intelligent?

Pervasive Intelligent Computing

Everywhere and Everytime

Pervasive Computing

- Nowadays the technology is highly pervasive
- We find “technological objects” everywhere and everytime
- There are “smart” meaning that they provide multiple services and are always “connected” and “interconnected”
- In many cases there are “intelligent”

Everywhere and Everytime

Some Case-Studies

- Smartphones
- Domotics
- Chatbots
- Smart Cars
- Humanoids

Smartphones!!

The Toys of 21th Century

- Calling...
- Playing...
- Browsing...
- Buying...
- Emailing...
- Chatting...
- Shooting...
- ...

Smartphones

Smartphones & Intelligence

- Fingerprint Recognition
- Navigation System
- Speech-to-Text
- “OK Google”
- Camera functions: autofocus, face recognition, red-eye elimination, ...

Smartphones

Smartphones & Intelligence

- **Fingerprint Recognition**
Image processing, minutiae identification
- **Navigation System**
Optimization algorithms
- **Speech-to-Text**
Neural networks
- **“OK Google”/Siri**
Text/lexical analyzers
- **Camera functions: autofocus, face recognition, red-eye elimination, ...**
Image Processing, Neural networks

Smart Objects and Internet-of-Things: Alexa, Google-Home

- We can ask everything...
- To control our home!
- News
- Music
- Weather
- Wake-up calls
- ...



Smart Objects and Internet-of-Things: Alexa, Google-Home

- **Speech-to-Text**
Neural networks
- **Chatbot**
Text processor/lexical analyzers



Chatbots, two main cases: Amazon and Vodafone

- While we understand that Alexa is not a real person...
- ...who is behind the scenes when you chat with Amazon or Tobi?
- It is a chatbot (a software robot) that tries to understand the question and...
- ...tries to reply with meaningful phrases by querying a proper knowledge base
- Text/Lexical Analyzers, Knowledge-Based Systems

Humanoids

The Robot becomes “Person”

- The Boston Dynamics case
- Robots able to walk in hard environments
- They walk, run, jump, go upstairs and downstairs, etc.
- Reinforcement Learning



Smart Cars

Smart cars & Intelligence

- Different Degrees of Smartness

Navigation System

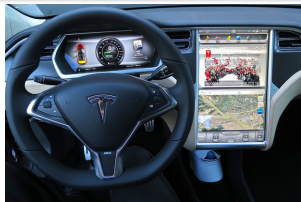
Cruise/Speed Limit Control

Obstacle recognition

Lane detector

Autonomous Parking

Autonomous Drive



Artificial Intelligence

Artificial Intelligence

Two Main Paradigms

- **Symbolism**
- **Connectionism**

Symbolism

- Represent the world by means of **symbols**
- Reason by manipulating these **symbols**

Connectionism

- Represent the world by means of **small interconnected entities**
- Reason by making entites **evolve and interact**

Symbolic Approaches

Knowledge Based/Expert Systems

- Strong use of the first-order logic (FOL)
- Map your world by means of FOL predicates
- Define your basic knowledge with some predicates that are always true (facts)
- Use logic clauses to infer new knowledge and/or determine actions

modem(off) $\Rightarrow$ turn_modem(on)

modem(on) $\wedge$ asdl_light(off) $\Rightarrow$ check_adsl_cable()

modem(on) $\wedge$ asdl_light(off) $\wedge$ adsl_cable(ok) $\Rightarrow$ check_carrier()

Symbolic Approaches

The Closed-World Assumption

- They operate in a world with well defined boundaries
- Any concept/case that cannot be directly mapped to a fact or rule cannot be handled at all

... Opening the World

- ... unless you are able to construct at runtime a **new case**
- But you need a programming language able to support **self-modification**

Symbolic Approaches

Opening the World and Self-Modifying Programs

- LISP (LISt Processing) was the first programming language with the ability of self-modifying
- In LISP program and data are syntactically expressed in the same way: **lists of terms**, where a term can be a symbol, a numeric type, a string, another list, etc.
- e.g. “**(modem on)**” can be interpreted as a **fact**, meaning that the modem is ON, or, if evaluated, is a call to function **modem** with **on** as parameter

Symbolic Approaches and NLP

Natural Language Processing

- Symbolic approaches are often used together with NLP techniques
- The aim is to try to extract a meaning from a sentence
- If the sentence is an assertion, e.g. “Corrado’s car is gray”, a new knowledge can be derived:
 $has(corrado, car), color(car, corrado, gray)$
- If the sentence is a question, it can become a clause that is evaluated, e.g. “Who owns gray cars?” becomes:
 $\forall X : has(X, car) \wedge color(car, X, gray)$

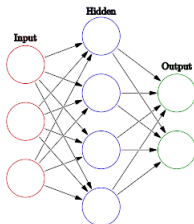
Chatbots

- Some chatbots use symbolic reasoning
- A **NLP** block performs a syntactical analysis of the sentence, thus deriving facts or primitive concepts
- A **reasoner**, with a more or less large set of rules, tries to understand the sentence (semantic analysis) in order to provide the right answer
- An knowledge base created a-priori is needed, populated with all the concepts of the domain of discourse

Connection-based Approaches

Neural Networks

- Optimization algorithms (gradient-descent) that uses a set of interconnected **artificial neurons**
- An **artificial neuron** is a piece of software that emulates the behaviour of a biological neuron
- The aim is to emulate the behaviour of biological neural systems



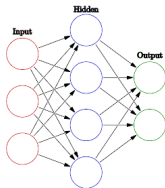
Connection-based Approaches

Neural Networks: Learning Phase

- They work by “crunching numbers” and “by examples”
- By continuously presenting couples (*input*, *output*) the algorithm modifies connections in order to determine the best function $f(\cdot)$ that approximates $output = f(input)$

Neural Networks: Evaluation Phase

- Then, by presenting a new (unknown) *input*, the network determines the output according to the optimized function $f(\cdot)$ above



Connection-based Approaches

Neural Networks and Keyword Spotting

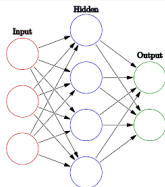
- If I want to make a system recognize the words “**Stop**” and “**Go**”, I can give audio samples of both words (learning phase):

input = samples_of_Stop, output1 = 1, output2 = 0

input = samples_of_Go, output1 = 0, output2 = 1

input = samples_of_Other_Words, output1 = 0, output2 = 0

- Then by presenting a **new** set of audio samples, the network will “activate” (output close to 1) one of the outputs



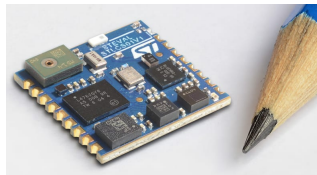
Connection-based Approaches

Neural Networks and Bad News

- And if the audio samples come from the word “Hello”?
- The algorithm is **hard**, the learning phase can even last **days**
- This led to “winter” in the research on NN, till now...

Neural Networks and Reinassance

- **Deep Learning**, now possible thanks to new high performance hardware (GPUs)
- **Neural IoT Systems**, now possible thanks to high performance MCU

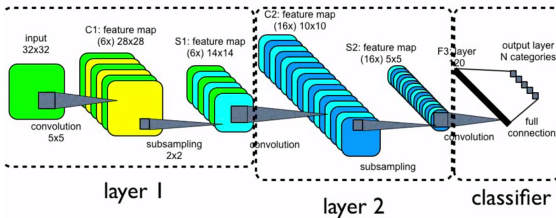


Connection-based Approaches

Neural Networks in our daily life

- Keyword Spotting
- Speech-to-text
- Image processing (classification, feature recognition, object recognition, etc.)

Convolutional Neural Networks



Other Kind of Learning-based System

Reinforcement Learning

- Boston Dynamics robots use Reinforcement Learning
- It is a learning technique that aims at finding the best solution by obtaining a “reward”
- The system generates a wide set of possible control action plans
- Each plan is executed and the outcome is determined
- If the outcome is “quite near” the goal, the plan receives a “reward”

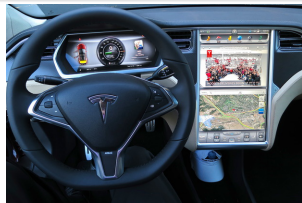


Planning

Smart Cars

Smart cars & Intelligence

- Navigation System: **Optimization Algorithms**
- Cruise/Speed Limit Control: **Control Algorithms**
- Obstacle recognition: **Reactive Programs**
- Lane detector: **Image Processing + Control Algorithms**
- Autonomous Parking: **Trajectory following (Control Algorithms)**
- **Autonomous Drive: Planning**

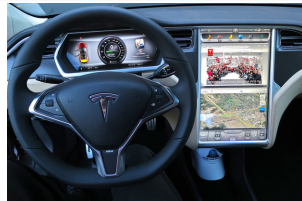


Planning

Autonomous Intelligent Systems (Agents)

The **Deliberation** process:

- 1 Given a certain **goal**
- 2 **Determine** the **possible plans** that could led to goal **achievement**
- 3 **Select** one of these plans, on the basis of a certain criteria
- 4 **Execute** the plan
- 5 **Check** if the plan succeeded
- 6 **Check** if the goal has been achieved, if not go to step 2

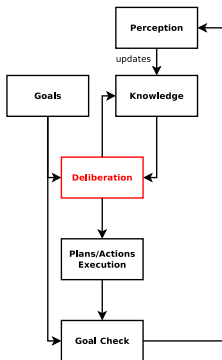


Planning

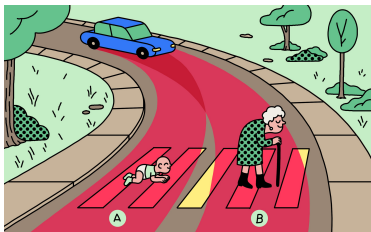
The BDI Paradigm

BDI: Belief-Desire-Intention

- **Beliefs:** the knowledge
- **Desires:** the goal(s)
- **Intentions:** the plans the agent intend to execute to achieve the goal(s)



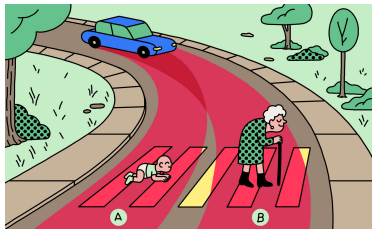
Intelligent Autonomous Systems: the “Trolley Problem”



- The first formulation is by Isaac Asimov
- **What should be the behaviour of the autonomous car?**
 - 1 **Stop suddenly:** the driver is hurt
 - 2 **Hit the baby:** the baby is hurt
 - 3 **Hit the lady:** the lady is hurt
 - 4 **Go off-road:** the driver is hurt

Any solution provokes a damage to a human being!!

Intelligent Autonomous Systems: the “Trolley Problem”



- In the case of an **accident**, who is responsible?
 - 1 The driver?
 - 2 The car manufacturer?
 - 3 The car itself?

In Summary...

In Summary

In Summary...

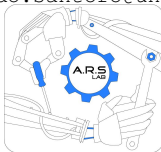
Are artificial system “intelligent”?

- Once again the question could be, what is the “intelligence”?
- We are quite far from “androids” like C-3PO
- But undoubtedly many algorithms “exhibit” an “rational behaviour”
- Currently we are able to **mimic** only some (small) aspects of human thinking and behaving
- No **“Terminator”** risk, but ...

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